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# Guided Individual Studies - Report

## Towards Identifiable Latent Dynamics: Integrating Sparse Model Discovery with Dynamics Contrastive Learning

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## Abstract

Nonlinear system identification aims to recover both an underlying low-dimensional state representation and the governing equations from high-dimensional measurements. A new algorithm called Dynamics Contrastive Learning (DCL) provides identifiability guarantees for the learned representation, but it does not yield explicit nonlinear governing equations, whereas Autoencoder-SINDy (AE-SINDy) targets interpretable sparse dynamics, but lacks comparable guarantees for representation recovery. This report evaluates AE-SINDy as a method for discovering low-dimensional, interpretable dynamics from high-dimensional observations of the Lorenz system. We first reproduce the original AE-SINDy Lorenz setup and show that its observation model is unusually benign, as even simple linear baselines (PCA/ICA) recover the latent trajectory almost perfectly after affine alignment. We then test AE-SINDy under a more realistic nonlinear mixing scheme with numerical approximations of the time derivatives computed from noisy observations. We show that in this setting, AE-SINDy no longer recovers the ground-truth latent variables reliably, and its alignment accuracy is comparable to PCA. Our results suggest that AE-SINDy can yield interpretable sparse dynamics when the latent representation is easy to recover, but that representation learning becomes a bottleneck under nonlinear mixing, motivating future work on combining identifiable representation learning (specifically via DCL) with sparse model discovery.

## CONTENTS

|            |                                                                       |    |
|------------|-----------------------------------------------------------------------|----|
| <b>I</b>   | <b>Introduction</b>                                                   | 1  |
| <b>II</b>  | <b>Background</b>                                                     | 2  |
| II-A       | System Identification . . . . .                                       | 2  |
| II-B       | Dynamics Contrastive Learning . . . . .                               | 3  |
| II-C       | The AE-SINDy Method . . . . .                                         | 4  |
| <b>III</b> | <b>Results</b>                                                        | 5  |
| III-A      | Assessment of the Original AE-SINDy Experimental Setup . . . . .      | 6  |
| III-B      | AE-SINDy Under Nonlinear Mixing and Approximate Derivatives . . . . . | 7  |
| III-C      | Consistency of AE-SINDy . . . . .                                     | 8  |
| <b>IV</b>  | <b>Discussion</b>                                                     | 9  |
| IV-A       | Future Directions . . . . .                                           | 9  |
| <b>V</b>   | <b>Conclusions</b>                                                    | 9  |
|            | <b>References</b>                                                     | 10 |

## I. INTRODUCTION

In this semester, I have worked with the group of Dr. Steffen Schneider at Helmholtz Munich. The group aims to develop machine learning algorithms for representation learning and nonlinear system identification, with applications for example in the life sciences. In their recent work [1], they developed a contrastive learning algorithm called *Dynamics Contrastive Learning* (DCL) that performs nonlinear system identification. However, while DCL learns latent representations with theoretical identifiability guarantees, it models nonlinear dynamics using switching linear dynamics and therefore does not recover explicit nonlinear governing equations.

To address this limitation, the group aims to use the *sparse identification of nonlinear dynamics* (SINDy) algorithm [2]. Accordingly, my work in the semester has focused on understanding DCL and Autoencoder-SINDy (AE-SINDy), the latter being an extension of SINDy introduced by Champion et al. [3]. Beyond merely familiarizing myself with the underlying concepts and methods, I also evaluated the performance of AE-SINDy under multiple conditions.

In this document, I first introduce the theoretical background necessary to understand the project and then present the preliminary results obtained so far.

## II. BACKGROUND

Data-driven discovery of governing equations is a long-sought goal of science and engineering [3]. Although the amount of data available for model inference continues to increase, discovering models that simultaneously satisfy three key properties — (i) accurate predictions, (ii) structural identifiability, and (iii) interpretability — remains challenging [2]. Models that provide arbitrarily accurate predictions, such as neural networks, are typically too complex to be interpretable and often contain too many parameters to be identifiable [4], [3], [2]. Simple and parsimonious models, on the other hand that recover the true underlying dynamics are difficult to discover purely from data without explicit domain knowledge such as the right coordinate system or the appropriate model structure [2].

In the following sections, we briefly introduce and define the notion of *structural identifiability*, and discuss two distinct approaches from the literature (Champion et al. [3] and Laiz et al. [1]) that aim to learn low-dimensional representations of dynamical systems. For each approach, we outline both its strengths and its limitations.

### A. System Identification

The aim of system identification is to uncover a mathematical model that describes the input–output relationship of a system [5]. In practice, this involves identifying both a model structure (i.e., the form of the governing equations) and the corresponding parameters [6]. This, however, is not always possible, as different model structures or parameter values may produce identical input–output behavior. The concept of identifiability therefore addresses whether, given observed measurement data (and possibly known inputs), the parameters of a mathematical model can be uniquely determined from finite data by solving the associated inverse problem [7].

In general, given a system operator  $S$  that maps system inputs to outputs under a parameter vector  $\mathbf{p}$ , the property of (structural) identifiability is defined as follows [7]:

$$\forall(\mathbf{p}, \mathbf{p}') : S[\mathbf{p}, u] = S[\mathbf{p}', u] \implies \mathbf{p} = \mathbf{p}'. \quad (1)$$

That is, if two parameter vectors  $\mathbf{p}$  and  $\mathbf{p}'$  produce identical outputs under the same input  $u$ , then the parameters must be equal. A simple violation of this condition can be seen by considering the system  $y = (m_1 \cdot m_2)x$ . In this case, the parameter vector  $\mathbf{p} = [m_1, m_2]^T$  is not identifiable from measurements, since the equality  $m_1 m_2 = m'_1 m'_2$  does not imply  $m_1 = m'_1$  and  $m_2 = m'_2$ .

The general definition in (1) can be made more explicit for the case considered here. We assume that high-dimensional observed data  $y \in \mathbb{R}^n$  (e.g., neural recordings) are generated by a set of latent variables  $x \in \mathbb{R}^d$  according to a discrete-time dynamical system of the form

$$x_{t+1} = f(x_t) + Bu_t + \varepsilon_t; \quad y_t = g(x_t) + \nu_t, \quad (2)$$

where  $u_t \in \mathbb{R}^d$  is a control signal whose effect is mediated by  $B \in \mathbb{R}^{d \times d}$  [1]. The function  $f : \mathbb{R}^d \mapsto \mathbb{R}^d$  describes the latent dynamics, while the nonlinear and injective mixing function  $g : \mathbb{R}^d \mapsto \mathbb{R}^n$  maps latent states to the observation space. The terms  $\varepsilon_t \in \mathbb{R}^d$  and  $\nu_t \in \mathbb{R}^n$  represent process and observation noise, respectively. The goal of nonlinear system identification in this setting is to identify the functions  $f$  and  $g$  from observations of  $y_t$ , and thereby recover the inverse of  $g$ , the so-called *de-mixing* function  $h : \mathbb{R}^n \mapsto \mathbb{R}^d$  [1].

This task essentially reduces to a representation learning problem. Representation learning aims to find lower-dimensional representations of data from which relevant features can be more easily extracted for prediction and inference [8], [9]. In the present context, this means learning functions  $h$  and  $f$  such that observed data are mapped to a low-dimensional latent space in which the system dynamics are accurately described and predictive of future observations. This task can be addressed using variational autoencoders and other deep neural network architectures, which are powerful function approximators due to their large number of parameters [9], [4].

Khemakhem et al. showed that recovering the true joint distribution of latent variables and observations requires explicit identifiability conditions on the model, even in the variational autoencoder framework [10]. In a related line of work, Hyvärinen et al. [11], and later Schneider et al. [12], [13], demonstrated that contrastive learning based on temporal data can enable representation learning with theoretical identifiability guarantees, up to known indeterminacies such as linear or affine transformations.

In the next section, I introduce the self-supervised contrastive learning approach developed by Schneider's group, which achieves nonlinear system identification with theoretical guarantees. The results of this method for the Lorenz system, as well as its limitations, are also discussed.

### B. Dynamics Contrastive Learning

Contrastive learning is a representation learning approach that trains a model to bring positive (similar) samples closer together and push dissimilar (negative) samples farther apart in an embedding space [1]. Its core idea is to learn meaningful features by comparing samples to one another, rather than relying on explicit labels.

In their work, Laiz et al. [1] build on previous work by Schneider et al. [12], who propose a contrastive learning approach to model time-course data. In this framework, the current observation  $y_t$  is assigned as the anchor ( $y$ ), the subsequent observation  $y_{t+1}$  as a positive sample ( $y^+$ ), and multiple observations  $y_i^- \in \mathcal{N}$  (with  $i \neq t$  and  $i \neq t+1$ ) as negative samples. A contrastive loss in CEBRA encourages the latent variables  $x_t$  and  $x_{t+1}$  corresponding to  $y_t$  and  $y_{t+1}$  to be close in the latent space.

DCL refines this objective by requiring not merely that  $x_t$  and  $x_{t+1}$  are close to each other, but that the prediction  $f(x_t)$  of the learned dynamics model  $f$  is close to  $x_{t+1}$ . The resulting DCL model  $\psi$  is specified by the log-likelihood given in Eq. (3).

$$\log p_\psi(y | y^+, \mathcal{N}) = \psi(y, y^+) - \log \sum_{y' \in \mathcal{N} \cup \{y^+\}} \exp(\psi(y, y')). \quad (3)$$

The full model  $\psi$  is trained by minimizing the negative log-likelihood over time-series data (formula not shown here) and is parameterized to ensure favorable identifiability properties. Based on the generative model in Eq. (2),  $\psi$  is defined as the composition of a de-mixing function  $h : \mathbb{R}^n \mapsto \mathbb{R}^d$ , an estimated dynamics model  $\hat{f} : \mathbb{R}^d \mapsto \mathbb{R}^d$ , a similarity function  $\phi : \mathbb{R}^d \times \mathbb{R}^d \mapsto \mathbb{R}$ , and a correction term  $\alpha : \mathbb{R}^d \mapsto \mathbb{R}$ . Formally,

$$\psi(y, y') = \phi(\hat{f}(h(y)), h(y')) - \alpha(y'). \quad (4)$$

The resulting algorithm, termed *Dynamics Contrastive Learning* (DCL), takes a pair of observations  $(y, y')$ , maps them into a latent space via  $h(y) = \mathbf{L}x + b$  and  $h(y') = \mathbf{L}x' + b$ , applies the learned dynamics model  $\hat{f}$ , and evaluates similarity using  $\phi$ . The overall procedure is illustrated in Figure 1a.

Laiz et al. study a restricted version of the generative model in Eq. (2) by setting  $\mathbf{B} = 0$  and  $\nu_t = 0$ . Their most relevant contribution for the present work is an identifiability result. Specifically, Theorem 1 in [1] states that, given a time-series dataset  $\{y_t\}_{t=1}^T$  generated by a bijective dynamics function  $f : \mathbb{R}^d \mapsto \mathbb{R}^d$  and an injective mixing function  $g : \mathbb{R}^d \mapsto \mathbb{R}^n$  as defined in Eq. (2), and under additional technical assumptions (including  $\varepsilon_t \sim \mathcal{N}(0, \Sigma_\varepsilon)$ ), the learned representation becomes identifiable in the limit of  $T \rightarrow \infty$ .

More precisely, for any latent state  $x$  occurring with nonzero probability, the composition of the learned encoder and the true mixing function satisfies  $h(g(x)) = \mathbf{L}x + b$  recovering the true latent variable up to an affine transformation. Furthermore, the estimated dynamics  $\hat{f}$  are bijective and obey  $\hat{f}(x) = \mathbf{L}f(\mathbf{L}^{-1}(x - b)) + b$ , implying that  $\hat{f}$  recovers the true dynamics  $f$  up to the same affine transformation determined by  $\mathbf{L}$ . A graphical summary of these results is shown in Fig. 1b.

Laiz et al. further evaluate their algorithm on systems exhibiting linear, switching linear, and nonlinear dynamics. In each case, they generate higher-dimensional observations from low-dimensional latent dynamics using a fixed 4-layer MLP followed by a linear mapping to a 50-dimensional observation space. They demonstrate in Figure 5 of their paper that DCL successfully identifies the dynamics of the chaotic Lorenz system when additive Gaussian noise with standard deviation  $\sigma = 0.001$  is present.

In addition to accurately modeling the dynamics, DCL produces latent representations of the observed data that are consistent across independent training runs. Here, consistency refers to the fact that the learned latent embeddings differ only by an affine transformation across runs [12]. As a result, DCL effectively addresses the problem of identifiable latent representations.

However, an important limitation remains. DCL approximates nonlinear dynamics using switching linear models, which means that (i) it does not always explicitly recover the true nonlinear governing equations and (ii) it does not yield an interpretable, closed-form description of the underlying dynamical system. These limitations are precisely the aspects addressed by the AE-SINDy algorithm, which aims to recover

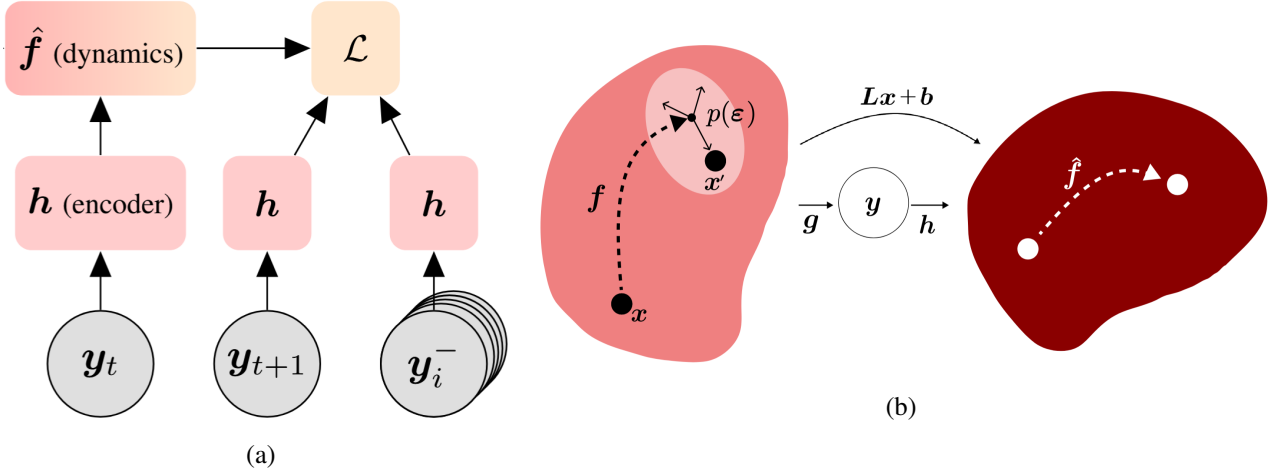


Figure 1: Dynamics Contrastive Learning framework and identifiability result. (a) Algorithmic structure of DCL. Pairs of observations  $(y_t, y_{t+1})$ , together with negative samples  $y_i^-$ , are mapped into a latent space by a shared de-mixing function  $h$ . A learned dynamics model  $\hat{f}$  propagates the latent representation forward in time, and the model is trained using the InfoNCE loss ( $\mathcal{L}$ ) based on a similarity measure  $\phi$ . (b) Graphical illustration of the identifiability result. Observations  $y$  are generated from ground-truth latent variables  $x$  (lighter-colored space, left) through an injective mixing function  $g$ . The learned encoder  $h$  approximately inverts the mixing and maps observations into a predicted latent space (darker-colored space, right). The true dynamics  $f$  propagate the latent state forward in time from  $x$  to  $x'$ , with additive noise  $\varepsilon$ . Under the assumptions of Theorem 1, both the estimated dynamics  $\hat{f}$  and the composition  $h \circ g$  recover the true dynamics and latent representation up to an affine transformation. The figure is reproduced and borrowed from the work of Laiz et al. [1].

sparse and interpretable governing equations, albeit - as we will show here - without strong guarantees of representation consistency.

In the next section, we introduce the AE-SINDy framework and discuss how the complementary strengths and weaknesses of DCL and AE-SINDy motivate their combination as a potential route toward both identifiable representations and interpretable dynamical models.

### C. The AE-SINDy Method

The AE-SINDy algorithm, introduced by Champion et al. [3], uses an autoencoder to discover a low-dimensional latent representation from high-dimensional observed data and subsequently applies the SINDy algorithm [2] to identify a sparse model of the dynamics in this learned latent space. In this sense, AE-SINDy can be viewed as wrapping an autoencoder around the SINDy framework. Accordingly, we first describe the SINDy method and then briefly summarize how it is combined with an autoencoder in AE-SINDy.

SINDy, short for *sparse identification of nonlinear dynamics*, formulates nonlinear model discovery as a sparse regression problem. Given a general nonlinear dynamical system  $\dot{x} = f(x, t)$ , SINDy constructs three data matrices, shown in Eq. (5):

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}^\top(t_1) \\ \mathbf{x}^\top(t_2) \\ \vdots \\ \mathbf{x}^\top(t_m) \end{bmatrix}, \quad \dot{\mathbf{X}} = \begin{bmatrix} \dot{\mathbf{x}}^\top(t_1) \\ \dot{\mathbf{x}}^\top(t_2) \\ \vdots \\ \dot{\mathbf{x}}^\top(t_m) \end{bmatrix}, \quad \Theta(\mathbf{X}) = \begin{bmatrix} | & | & | & | & \cdots & | & | & \cdots \\ \mathbf{1} & \mathbf{X} & \mathbf{X}^{\mathbf{p}_2} & \mathbf{X}^{\mathbf{p}_3} & \cdots & \sin(\mathbf{X}) & \cos(\mathbf{X}) & \cdots \\ | & | & | & | & & | & | & \end{bmatrix} \quad (5)$$

Here, each row of  $\mathbf{X} \in \mathbb{R}^{m \times d}$  and  $\dot{\mathbf{X}} \in \mathbb{R}^{m \times d}$  corresponds to a snapshot of the latent state  $\mathbf{x} \in \mathbb{R}^d$  and its time derivative at one of  $m$  time points. The matrix  $\Theta(\mathbf{X}) \in \mathbb{R}^{m \times p}$  defines a library of  $p$  candidate nonlinear functions, evaluated row-wise on the data, which serve as regressors in the sparse regression problem. Using this library, SINDy seeks to solve the following equation for  $\Xi$ :  $\dot{\mathbf{X}} = \Theta(\mathbf{X})\Xi$  where  $\Xi = [\xi_1, \xi_2, \dots, \xi_d] \in \mathbb{R}^{p \times d}$  is a sparse coefficient matrix whose columns correspond to the governing equations of the individual state variables.

In AE-SINDy, the coefficients in the  $\xi_i$  vectors are learned jointly with the parameters of the neural network. Sparsity is enforced through sequential thresholding, that is, after every  $i^{\text{th}}$  training iteration, all entries of the coefficient matrix  $\Xi$  whose absolute value falls below a fixed threshold (set to 0.1 in the original work, but treated as a tunable hyperparameter) are set to zero and remain fixed at 0 for the remainder of training session.

AE-SINDy employs a composite loss function consisting of four terms, each targeting a different aspect of the learned representation and dynamics. Given high-dimensional observations  $y \in \mathbb{R}^n$  generated from latent variables  $x \in \mathbb{R}^d$  via  $y_t = g(x_t)$ , the AE-SINDy loss is defined as

$$\underbrace{\|y - g(x)\|_2^2}_{\mathcal{L}_{\text{recon}}} + \lambda_1 \underbrace{\left\| \dot{y} - (\nabla_x g(x)) (\Theta(x^\top) \Xi) \right\|_2^2}_{\mathcal{L}_{dy/dt}} + \lambda_2 \underbrace{\left\| (\nabla_y x) \dot{y} - \Theta(x^\top) \Xi \right\|_2^2}_{\mathcal{L}_{dx/dt}} + \lambda_3 \underbrace{\|\Xi\|_1}_{\mathcal{L}_{\text{reg}}}, \quad (6)$$

where  $\mathcal{L}_{\text{recon}}$  is the autoencoder reconstruction loss,  $\mathcal{L}_{dy/dt}$  penalizes mismatches between the observed time derivative  $\dot{y}$  and the derivative predicted via the learned latent dynamics and decoder,  $\mathcal{L}_{dx/dt}$  enforces consistency between the true latent dynamics and the SINDy model, and  $\mathcal{L}_{\text{reg}}$  promotes sparsity in the coefficient matrix  $\Xi$ . Here,  $\nabla_x g(x)$  and  $\nabla_y x$  denote the gradients of the decoder and encoder, respectively. The relative importance of the individual loss terms is controlled by the hyperparameters  $\lambda_i$ , which are selected by trial and error in the original work.

One representative experiment considered by Champion et al. is the reconstruction of the Lorenz attractor from high-dimensional observations of the form

$$y_t = u_1 x_1(t) + u_2 x_2(t) + u_3 x_3(t) + u_4 x_1(t)^3 + u_5 x_2(t)^3 + u_6 x_3(t)^3 \in \mathbb{R}^{128}, \quad (7)$$

where,  $x = [x_1, x_2, x_3]^\top$  follows the Lorenz dynamics, and the vectors  $u_i$  correspond to the first six Legendre polynomials evaluated at 128 spatial grid points.

In their paper [3], Champion et al. report discovering several parsimonious models from the data defined above, with some reconstructions being exceptionally close to the true Lorenz attractor — up to an affine transformation. However, even across just ten runs of the same model initialized with different random weights, the recovered dynamics vary substantially as highlighted in the supplementary material of Champion et al. under the section of the Chaotic Lorenz System. Additionally, throughout their experiments, Champion et al. compute the time derivative of the observed variables  $y \in \mathbb{R}^n$  using analytically derived expressions propagated from the known latent dynamics, rather than estimating derivatives from the observed data itself. This setup bypasses a major practical difficulty of system identification and does not reflect realistic experimental conditions, where time derivatives must be approximated numerically from noisy measurements.

Therefore, in the next section, we replicate the Lorenz experiment originally performed by Champion et al. and show the benign nature of the data generation process defined in Eq. (7). Additionally, we also test AE-SINDy in a more realistic regime, where we approximate the time derivatives of  $y \in \mathbb{R}^n$  numerically and introduce a more complex nonlinear mixing function  $g: \mathbb{R}^d \mapsto \mathbb{R}^n$ . For this latter experiment, we also compute and present here the consistency across AE-SINDy models.

### III. RESULTS

To begin with, we replicated the AE-SINDy experiments using the same setup as in the original paper, following the authors' publicly available code [3]. The Lorenz system is simulated for  $m = 250$  time points and for 1024 and 20 different initial conditions randomly sampled for the training and validation sets, respectively, with additive observation noise of  $\sigma = 10^{-6}$ . This corresponds to a noise variance of  $10^{-12}$ . The high-dimensional observations  $y \in \mathbb{R}^{128}$  are generated according to Eq. (7). As in [3], we restrict the candidate function library  $\Theta(\mathbf{X})$  to polynomial terms only. During training, the data is passed through the AE-SINDy architecture with randomly initialized network weights, using different random seeds for each run. Training consists of 2100 standard epochs followed by 500 so-called refinement epochs, which are used to fine tune the entries of the  $\xi_i$  parameter vectors that have not been thresholded.

The first point at which our setup deviates from the code provided by Champion et al. concerns the epoch numbers. In their implementation, the network is trained for 5000 epochs followed by 1000 refinement epochs. However, as shown in Fig. 2, both the training and validation losses defined in Eq. (6) decrease rapidly during the first 100-200 epochs, followed by a much slower decrease before effectively saturating at around the 2000<sup>th</sup> iteration.

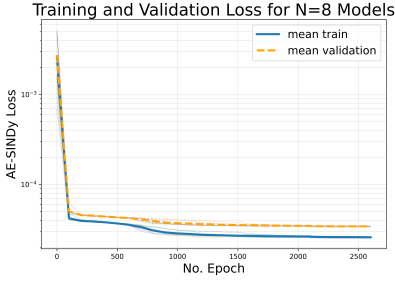


Figure 2: Training and validation loss curves for AE-SINDy averaged over  $N = 8$  runs. Solid lines show mean losses; faint lines show individual runs.

initialized at  $[x_1, x_2, x_3]^\top = [-8, 7, 27]^\top$  and simulated for  $m = 2000$  time points from  $t_1 = 0$  to  $t_m \approx 20$  with a time step of  $dt = 0.01$ . This results in 2000 samples in the test dataset. The latent embeddings produced by the autoencoder are extracted and subsequently aligned to the ground-truth (GT) latent variables using an affine transformation.

The curves shown in Fig. 2 represent mean losses over  $N = 8$  independent training runs, each initialized with a different random seed.

Based on these observations, all subsequent experiments presented here use 2000 training epochs followed by 500 refinement epochs. When using the exact same setup (except for the epoch numbers) as in [3], the code provided by Champion et al. replicates their results exactly and thus these are not displayed here. Instead, in the next section, we demonstrate that the data generation process defined in Eq. (7) and used in [3] is overly simplistic.

#### A. Assessment of the Original AE-SINDy Experimental Setup

We ran the AE-SINDy algorithm according to the original setup specified by [3], with the exception of the noise strength, which we set to  $\sigma = 10^{-3}$ . The autoencoder architecture is a 6-layer fully connected MLP with encoder  $d_{\text{in}} \rightarrow 64 \rightarrow 32 \rightarrow 3$  and symmetric decoder  $3 \rightarrow 32 \rightarrow 64 \rightarrow d_{\text{in}}$ , using sigmoid activations in the hidden layers and linear output layers. The model was evaluated on a single test trajectory

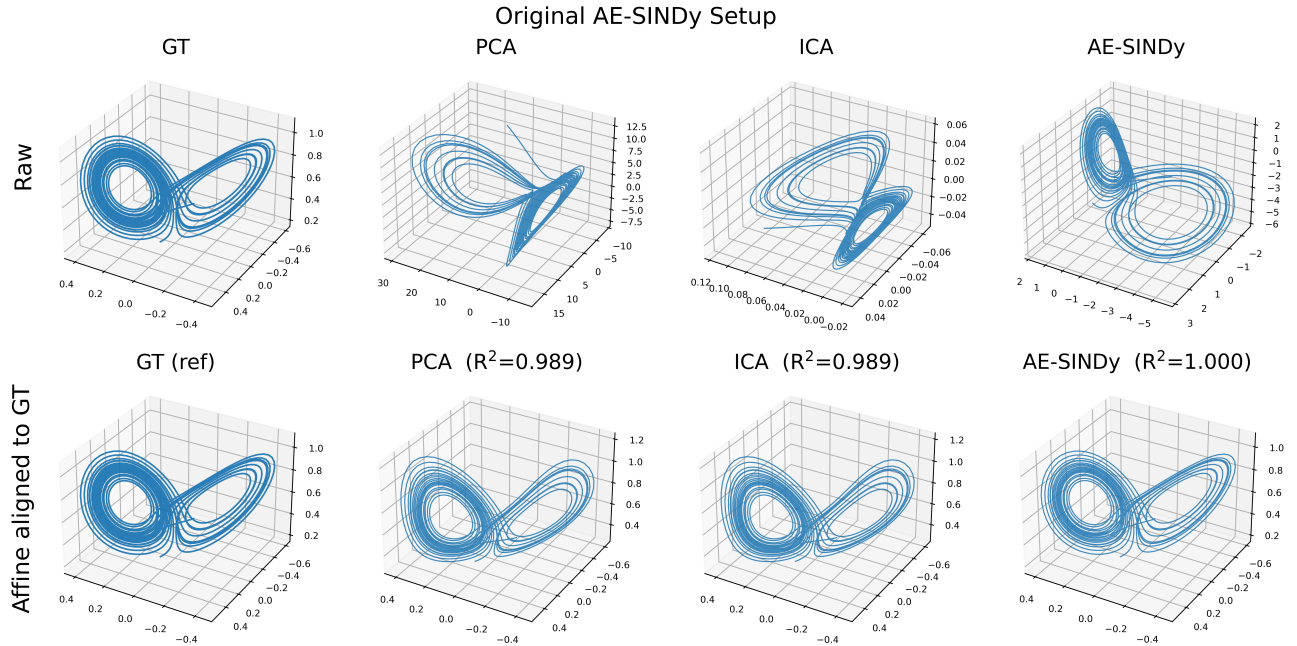


Figure 3: Raw and affine-aligned latent embeddings obtained under the original AE-SINDy experimental setup are shown. The top row shows the raw three-dimensional latent trajectories extracted from a single test trajectory, while the bottom row shows the same embeddings after affine alignment to the ground-truth (GT) latent trajectory. Columns correspond, from left to right, to the ground truth (GT), PCA, ICA, and AE-SINDy models. In the raw embeddings (top row), all three methods recover trajectories that resemble the geometric structure of the Lorenz attractor up to affine distortions in scale, orientation, and offset. After affine alignment, however, (bottom row), the PCA-, ICA-, and AE-SINDy-derived embeddings all closely coincide with the ground-truth trajectory. The coefficient of determination ( $R^2$ ) for each aligned method is reported in the subplot titles, demonstrating near-perfect recovery of the latent dynamics across all methods.

To assess whether recovering the true latent variables from the high-dimensional observations is inherently difficult under this setup, we additionally apply Principal Component Analysis (PCA) and Independent Component Analysis (ICA) directly to the observed data  $y \in \mathbb{R}^{128}$ . The resulting raw and affine-aligned latent trajectories are shown in Fig. 3.

Figure 3 shows that the raw latent embeddings recovered by PCA, ICA, and AE-SINDy already closely resemble the true Lorenz attractor up to simple geometric transformations. After applying an affine alignment, all three methods recover the ground-truth trajectory with near-perfect accuracy. This observation suggests that the data generation process defined in Eq. (7) produces unusually favorable conditions for latent recovery, even for purely linear methods such as PCA.

To evaluate how AE-SINDy performs in a more realistic setting — where the derivatives of the high-dimensional observations are not computed from analytic expressions but instead must be approximated from noisy data — we next introduce a modified data generation paradigm and assess AE-SINDy under these conditions.

### B. AE-SINDy Under Nonlinear Mixing and Approximate Derivatives

To evaluate the AE-SINDy algorithm in a more realistic scenario, we follow the data generation paradigm of Laiz et al. [1] and introduce a nonlinear mixing function  $g : \mathbb{R}^3 \mapsto \mathbb{R}^{50}$ . This mixing is defined as the composition of a fixed, randomly initialized 4-layer multi-layer perceptron (MLP) mapping from  $\mathbb{R}^3$  to  $\mathbb{R}^3$ , followed by a linear mapping  $\mathcal{A} : \mathbb{R}^3 \mapsto \mathbb{R}^{50}$ . The parameters of this mixing function are not learned and remain fixed throughout all experiments.

Additionally, instead of computing  $\dot{y} \in \mathbb{R}^{50}$  from analytic expressions of the underlying Lorenz system, we approximate the time derivatives numerically. Specifically, the observed signals are first smoothed using a Savitzky–Golay filter [14], [15], after which time derivatives are computed via a finite-difference scheme. We then apply the same evaluation pipeline as described in Section III-A, with the test trajectory now living in  $\mathbb{R}^{2000 \times 50}$  instead of  $\mathbb{R}^{2000 \times 128}$ . The resulting embeddings are shown in Fig. 4.

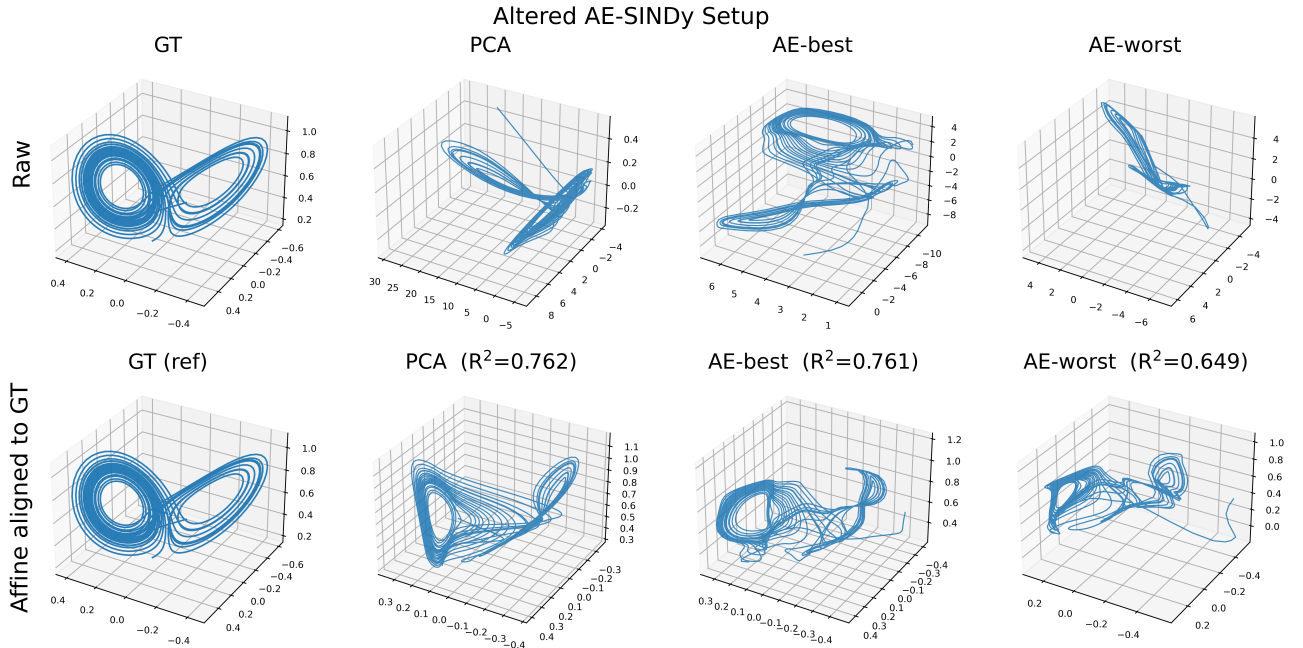


Figure 4: Raw and affine-aligned latent embeddings obtained under the altered AE-SINDy setup with nonlinear MLP mixing and approximate time derivatives are displayed. Each column corresponds to a single model, while rows distinguish between raw (top row) and affine-aligned (bottom row) embeddings. The first column shows the ground-truth latent Lorenz trajectory. The remaining columns show embeddings obtained via PCA and two representative AE-SINDy models selected from eight trained instances: the model achieving the highest affine alignment score (AE-best) and the model achieving the lowest one (AE-worst), as quantified by the coefficient of determination  $R^2$ . All affine alignments are performed with respect to the ground-truth latent trajectory. The raw embeddings already reveal substantial geometric distortions and (unlike in Fig. 3) these discrepancies persist after affine alignment. The reduced  $R^2$  values reported for the aligned embeddings further indicate a degraded recovery of the true latent dynamics compared to the original AE-SINDy setup in Fig. 3.

Figure 4 shows the latent embeddings before affine alignment (top row). Three approximations are shown alongside the ground-truth trajectory (blue). The AE-best and AE-worst labels refer to the AE-SINDy models

achieving the highest and lowest affine alignment scores, respectively, as quantified by the  $R^2$  metric. While differences in scale are expected and do not affect affine alignment, the recovered attractor geometries are visibly distorted relative to both the ground truth and the embeddings obtained under the original AE-SINDy setup.

This degradation becomes more apparent after affine alignment, as shown in the bottom row of Fig. 4. Compared to Fig. 3, the match between the recovered and true latent trajectories is substantially worse, which is reflected in the reduced  $R^2$  values.

In total, eight AE-SINDy models were trained under the nonlinear MLP mixing setup. For clarity, Fig. 4 displays only the best- and worst-performing models. However, affine alignment scores were computed for all eight runs. In addition, we examined the consistency of AE-SINDy under this nonlinear mixing regime with noise level  $\sigma = 10^{-3}$ . These results are presented in the following section.

### C. Consistency of AE-SINDy

The consistency scores for the different AE-SINDy models were computed using CEBRA, following the work of Schneider et al. [12], [13]. The consistency matrices in Fig. 5 display  $R^2$  values of affinely aligned latent embeddings across the eight different AE-SINDy models trained under the nonlinear MLP setup. Consistency scores for the original AE-SINDy setup, with the mixing function defined in Eq. (7), were also computed and in this case all embeddings achieved a consistency score of 100%. This is not surprising, as simple linear methods such as PCA and ICA were also able to recover the latent embeddings under this setup. All scores reported in Fig. 5 are expressed as percentages.

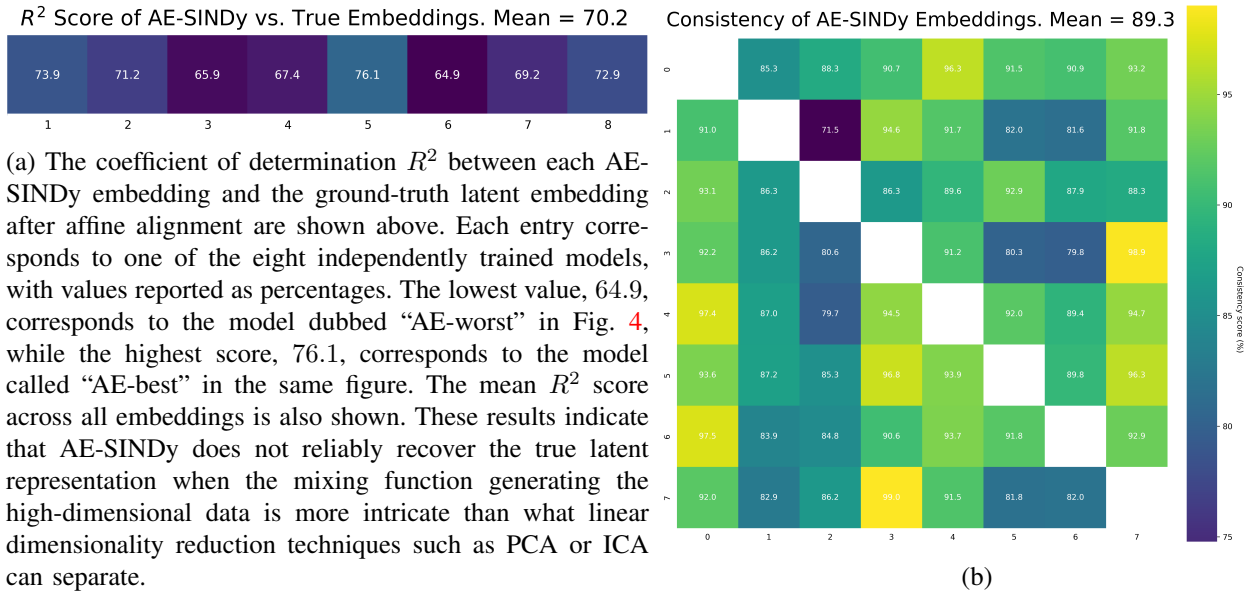


Figure 5: The Alignment accuracy and consistency of AE-SINDy latent embeddings obtained under the nonlinear MLP setup are displayed. Fig. 5a shows the  $R^2$  scores of each AE-SINDy embedding after affine alignment to the ground truth. Fig. 5b displays the pairwise consistency matrix of AE-SINDy embeddings computed using CEBRA. Each entry corresponds to the  $R^2$  value (in percent) obtained by affinely aligning the latent embedding of one model to that of another. Diagonal entries are omitted, as they correspond to self-alignment. The color scale encodes the magnitude of the consistency score.

In Fig. 5a, the  $R^2$  values of the affine alignments to the ground-truth latent embedding are shown for all eight AE-SINDy models. The scores, ranging from 64.9 to 76.1, demonstrate that none of the models accurately recover the true latent representation under the nonlinear MLP mixing setup.

Figure 5b shows the pairwise consistency matrix between AE-SINDy embeddings. Each entry represents the affine alignment between two independently trained models, with one embedding treated as the reference and the other as the target. This matrix therefore quantifies how consistently different AE-SINDy models represent the latent space, independent of the ground truth. The consistency scores range from 71.5 to 99.0, with a mean value of 89.3. Notably, this indicates that while AE-SINDy embeddings are relatively consistent across different runs, they are consistently misaligned with respect to the true latent variables.

#### IV. DISCUSSION

The results presented in Fig. 5 as well as those discussed in Section III-A indicate that while the consistency of AE-SINDy across runs is relatively strong, its ability to capture latent representations is essentially the same as that of a linear dimensionality reduction technique such as PCA. It should be noted, however, that AE-SINDy recovers a dynamical model for the latent variables in addition to the embedding. Therefore, even though its embedding capabilities are not superior to linear methods, in scenarios where low-dimensional representations of the data can be recovered easily, AE-SINDy can provide a latent embedding that is an affine transform away from the true one and, at the same time, yields an interpretable dynamical model of the nonlinear data generation process, as demonstrated in Fig. 3.

Furthermore, our PCA and ICA results in Section III-A suggest that Eq. (7) defines a mixing function that is easily invertible even by linear dimensionality reduction techniques such as PCA. This observation implies that the data generation process used in the original AE-SINDy experiments is overly optimistic. Consequently, to meaningfully evaluate the performance of AE-SINDy, it is necessary to test the algorithm on data generated through a clearly nonlinear mixing process.

We therefore proceeded to investigate the algorithm's performance under a more realistic nonlinear data generation scheme. Following Laiz et al. [1], a four-layer MLP was used as the nonlinear mixing function, with the noise level set to  $\sigma = 0.001$ , matching the conditions under which the DCL method is evaluated in [1]. This setup was chosen to assess how AE-SINDy recovers latent embeddings under a nonlinear observation model comparable to that used in the DCL framework. Our results reinforce the notion introduced above, namely that the AE-SINDy autoencoder behaves similarly to PCA or ICA when learning low-dimensional representations from high-dimensional data. In Fig. 4, this is reflected by the observation that  $R_{\text{PCA}}^2 \approx R_{\text{AE-SINDy}}^2$ .

To determine whether AE-SINDy's limitations are restricted to representation learning or extend to the consistency of the recovered embeddings, we further analyzed the consistency of the AE-SINDy models using the approach proposed by Schneider et al. [12]. The  $R^2$  scores and the consistency matrix shown in Figs. 5a and 5b, respectively, indicate that while AE-SINDy remains relatively consistent across different random initializations, it does not reliably recover the true low-dimensional latent representation of the Lorenz system under nonlinear mixing.

##### A. Future Directions

Our results ultimately suggest that AE-SINDy provides appropriate groundwork for the interpretable inference of governing equations of a potentially nonlinear data-generating process, but that it struggles with the task of representation learning. In particular, it fails to reliably identify the correct coordinate system in which the latent variables reside, and therefore can recover latent dynamics only in cases where the mixing function is sufficiently simple, in the sense for example, that even a linear method such as PCA can invert it.

As a consequence, we hypothesize that if AE-SINDy is combined with an architecture capable of performing representation learning in realistic scenarios where complex nonlinear mixing is present, then AE-SINDy can still serve as an effective framework for discovering low-dimensional, interpretable dynamics. In such a hybrid setup, the role of AE-SINDy would be to infer sparse and interpretable governing equations, once the appropriate latent coordinate system has been identified consistently by another method.

This hypothesis defines the direction of our future work and the broader potential of this project. Specifically, we aim to integrate AE-SINDy with Dynamics Contrastive Learning (DCL), using the latter to learn a consistent and identifiable low-dimensional latent representation across runs [1], while employing AE-SINDy and the SINDy framework to infer interpretable nonlinear dynamics within that representation.

#### V. CONCLUSIONS

In this project, we investigated the performance of the AE-SINDy framework for discovering latent dynamics from high-dimensional observations, with a particular focus on identifiability and consistency. While AE-SINDy reliably recovers sparse and interpretable dynamical models when the latent representation is simple, we showed that its representation learning capability is limited under realistic nonlinear mixing, where its performance is comparable to linear methods such as PCA. In contrast, approaches such as Dynamics Contrastive Learning, which we also discussed, provide identifiable and consistent latent

representations, but do not yield interpretable dynamical equations. These results motivate our future work on combining representation-learning methods with strong identifiability guarantees and sparse model discovery techniques, with the goal of jointly achieving consistency and interpretability in nonlinear system identification.

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